

Mexico city Household Travel Survey Comparison across Years (2007-2017)

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Overview

This analysis provides critical insight into how travel behaviors in Mexico City evolved between 2007 and 2017, particularly focusing on mode choice and the role of multimodal connectivity. Understanding these patterns is essential for transportation planning, as it highlights where high-capacity transit has gained or lost traction, how the private car's role is shifting, and how multimodal integration has expanded. These findings serve as a foundation for policy implication on improving transit connectivity, investing in high-capacity corridors, and reducing dependence on private automobiles.

Methodology

Defining trip types

We classified each trip record as either:

- **Single-mode:** the trip used exactly one transportation mode. (*Note: Without considering mode transfer, even within big categories, e.g., bus → metrobús → bus is not considering single-mode bus trip; but bus → bus is still considered single-mode bus trip*)
- **Multimodal:** the trip used two or more **distinct** modes.

To keep cross-year comparisons consistent, we **excluded walking legs when classifying multimodal trips** because walking transfers are often under-reported. Practically, this means we removed "walk" from a trip's mode list before classifying it. If a trip then had only one non-walk mode, it was treated as single-mode; if two or more non-walk modes remained, it was multimodal.

Because the 2007 survey did not record walking, **2017 walking trips are analyzed descriptively but excluded from percentage calculations used for 2007–2017 comparisons.**

De-duplication and order

Within a trip, we counted each mode **once**, regardless of repeats, and we ignored the **order** in which modes occurred.

- *Example (de-duplication):* bus → metrobús → bus is treated as **bus + metrobús** (not "bus" twice).
- *Example (order):* bus → metrobús and metrobús → bus are both classified as **bus + metrobús**.

Mode consolidation

To simplify categories, closely related sub-modes were merged into Metro, BRT, Non-BRT bus, and others (in multi-modes trips):

- **Tren + Metro → Metro** (fixed-guideway, high-capacity transit).

- **Bus + Minibus** → **Bus** (non-high-capacity bus services).
- **Drive+ bike+ motorcycle +etc.** → **others**

To facilitate comparison of multimodal patterns involving high-capacity transit, we combined BRT and Metro into a single high-capacity transit category.

Data processing and reproducibility

All processing and analysis were performed in **R**. The scripts and input data are available in the accompanying **GitHub repository** (packaged as **.zip** files).

Weighted vs Unweighted Comparison

There was little difference between the weighted and unweighted results in both 2007 and 2017. However, weighted estimates are preferred because they more accurately reflect the population and therefore provide a stronger basis for assessing mode shifts.

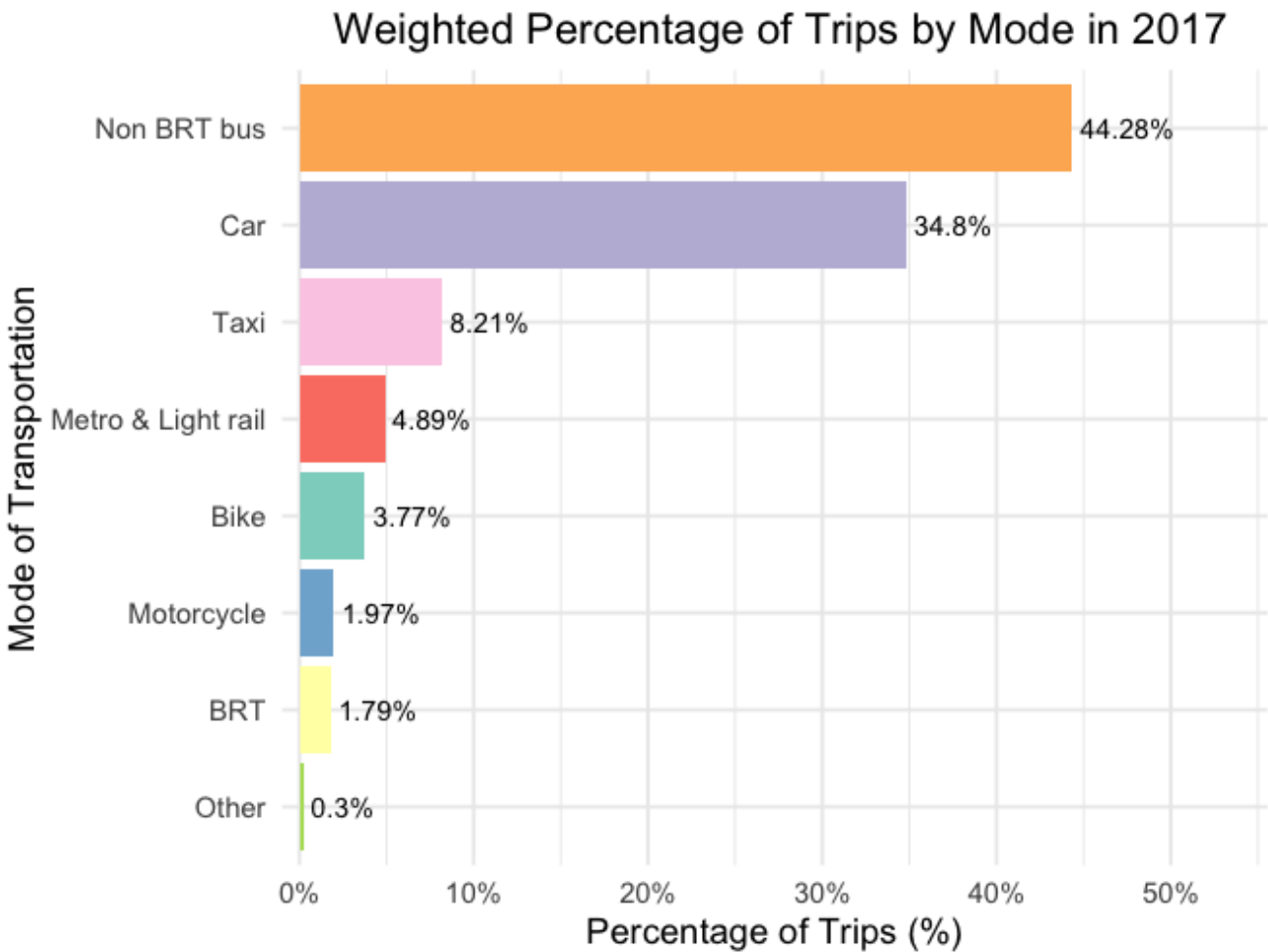
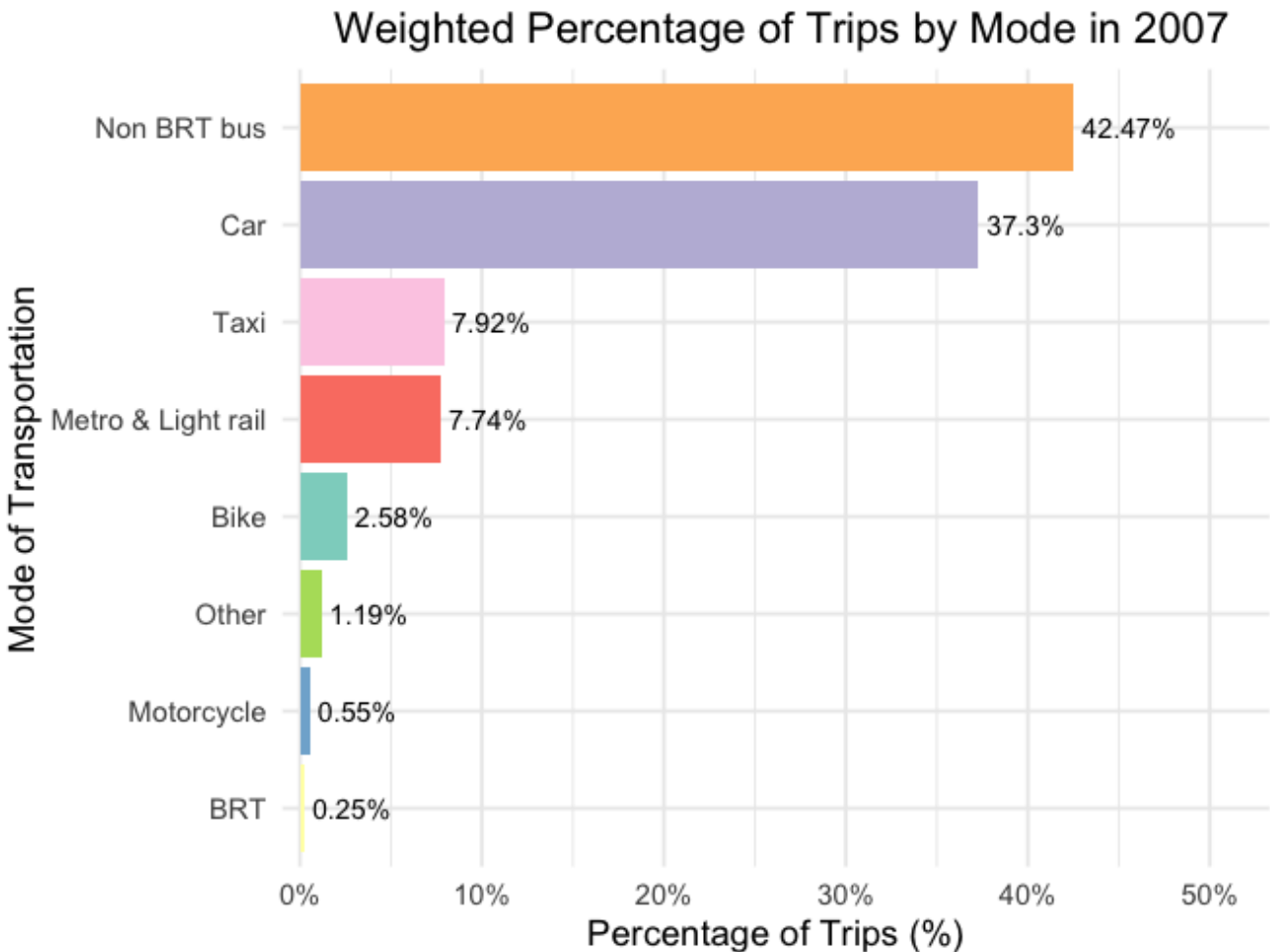
For single-mode trips, the weighted share was slightly lower in both years, with a more noticeable drop in 2017. Specifically, in 2017, weighted single-mode trips accounted for 79.25% (including walking), compared to 86.18% in the unweighted results.

All subsequent analyses in this report are based on the weighted results.

Single-mode trips trends (2007 vs 2017)

Single-mode trips remained dominant in both 2007 and 2017. Excluding walking, their share fell from 76.68% in 2007 to 70.47% in 2017; when walking is included, the 2017 single-mode share is 79.25%.

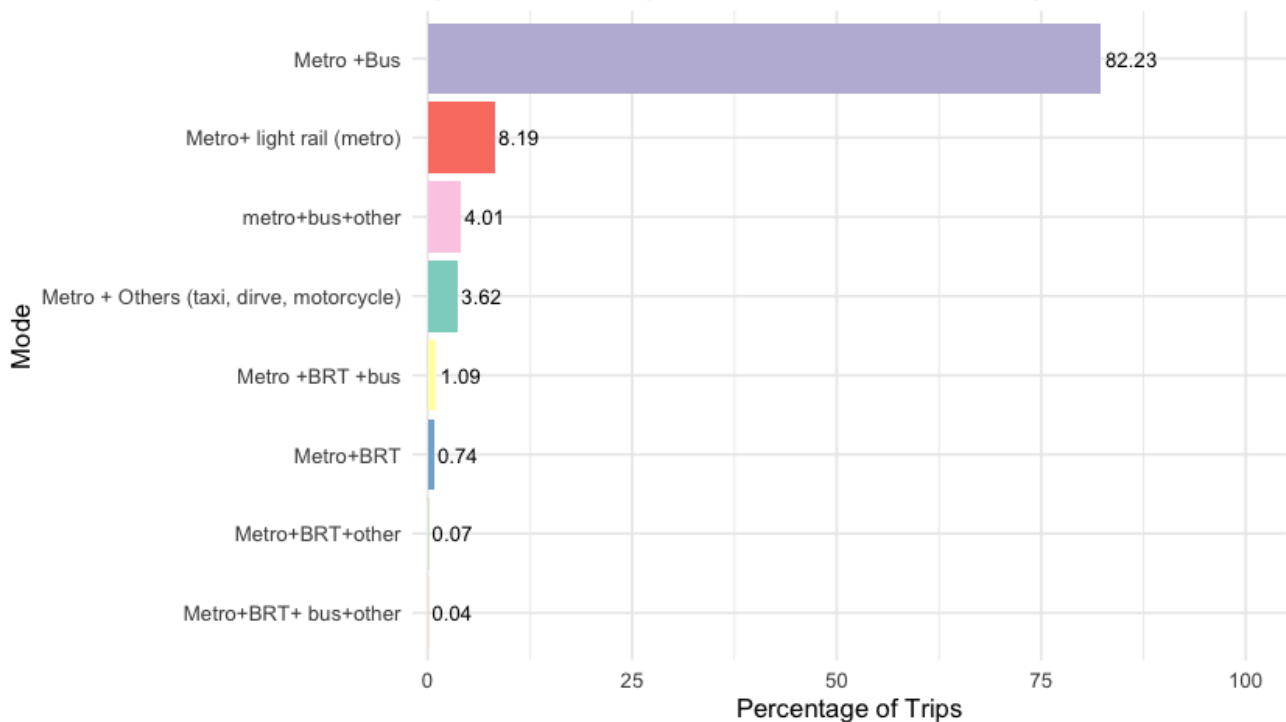
By mode, non-BRT bus was the most commonly used in both years, rising from 42.47% to 44.28%. Automobile (driver) declined from 37.3% to 34.8%, while taxi increased from 8% to 8.21%. Metro/light rail decreased from 7.74% to 4.89%, whereas BRT grew from 0.25% to 1.79%. Other modes (e.g., bike, motorcycle) remained at roughly 5% across both years.



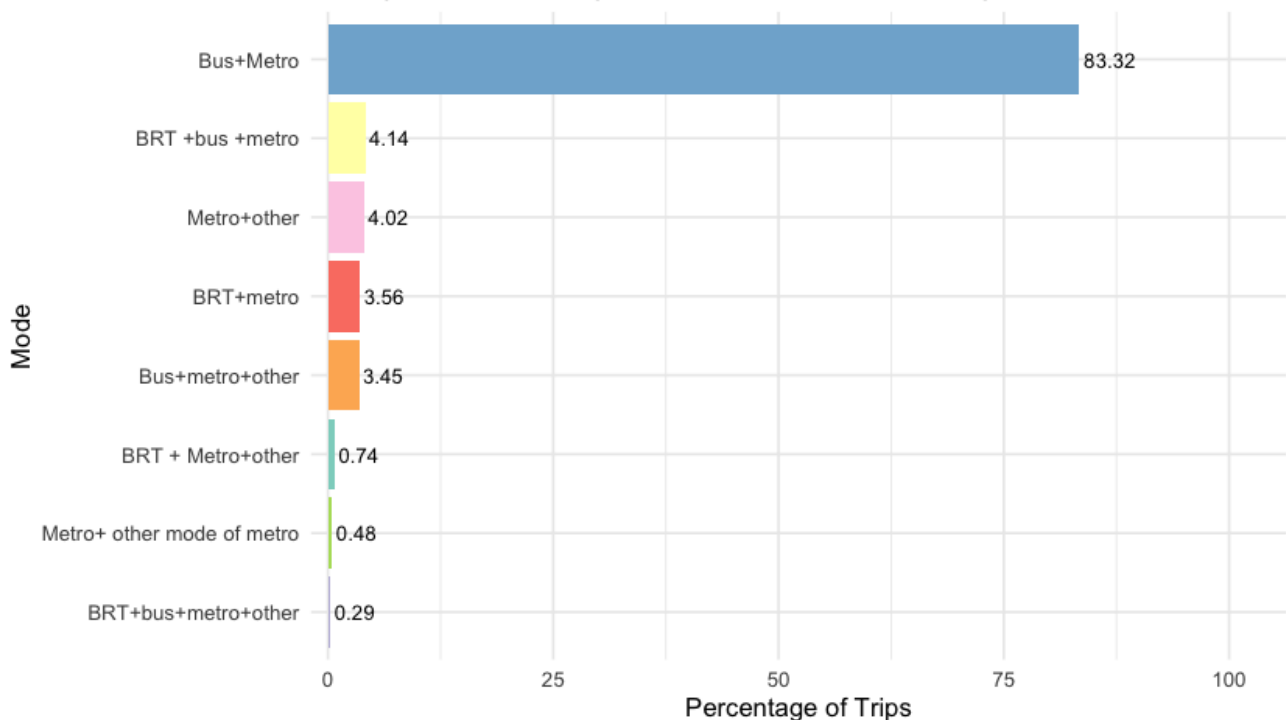
Multimodal trips trends (2007 vs 2017)

High-capacity transit (BRT and Metro) plays a more prominent role in multimodal trips than in single-mode travel. In single-mode trips, high-capacity modes account for 7.88% in 2007 and 5.62% in 2017. By contrast, multimodal trips that include Metro represent 19.66% of all trips in 2007 and 22.71% in 2017. Bus is the primary feeder to high-capacity transit: among Metro-involved multimodal trips, 82.23% (2007) and 83.32% (2017) are bus + Metro combinations.

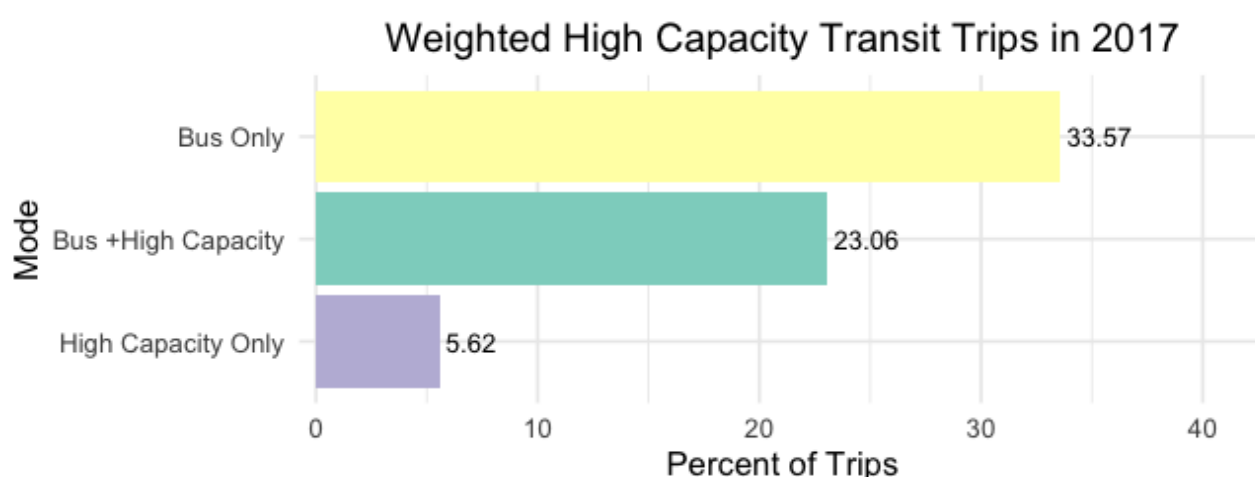
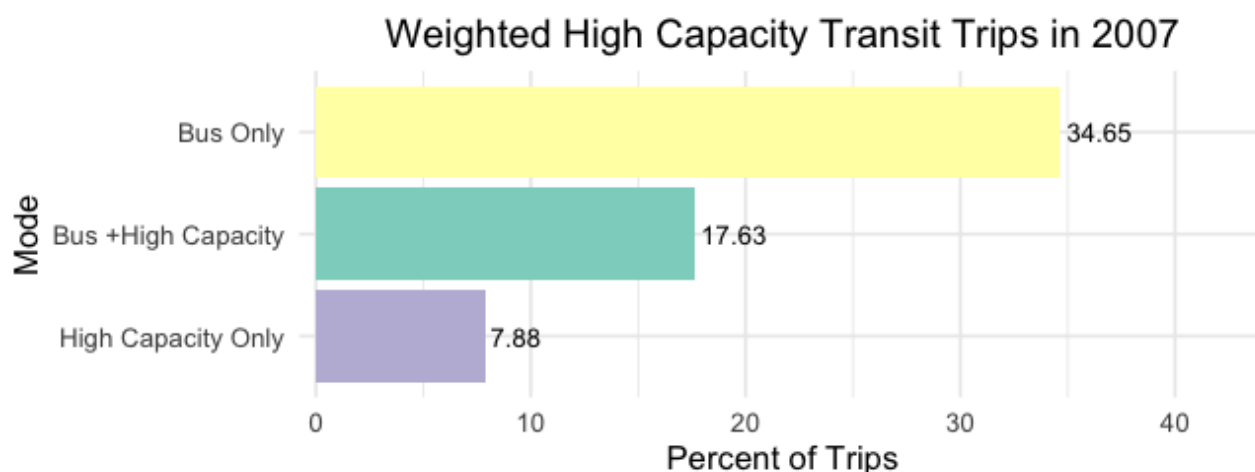
Weighted Percentage of Multimodal Metro & Light Rail Trips in 2007



Weighted Percentage of Multimodal Metro & Light Rail Trips in 2017



Overall, multimodality increased from 2007 to 2017. Bus-only trips decreased from 34.65% to 33.57%, and high-capacity-only (Metro/BRT) trips fell from 7.88% to 5.62%, while bus + high-capacity trips rose from 17.63% to 23.06%.



Key takeaways and insights

- **Single-mode travel remains dominant**, but its share declined over time. Excluding walking, single-mode trips dropped from 76.7% in 2007 to 70.5% in 2017.
- **Bus use stayed on top** as the largest share of single-mode trips, while automobile driving declined slightly, and taxis saw marginal growth. Metro use decreased in single-mode trips, though BRT showed notable growth.
- **Multimodality became more prevalent**, particularly combinations involving high-capacity transit. High-capacity-involved (BRT or Metro) multimodal trips grew from 20% in 2007 to 24% in 2017, with bus serving as the key feeder mode.
- **Connectivity between bus and high-capacity transit strengthened**, with bus + Metro/BRT trips rising from 17.6% to 23.1% of all trips, indicating a better integrated public transportation networks, especially in high-capacity+bus connectivity.
- **High-capacity transit is disproportionately important in multimodal contexts**, even as its share in single-mode trips fell. This suggests its primary role lies in enabling longer, more complex travel chains rather than acting as a stand-alone mode.

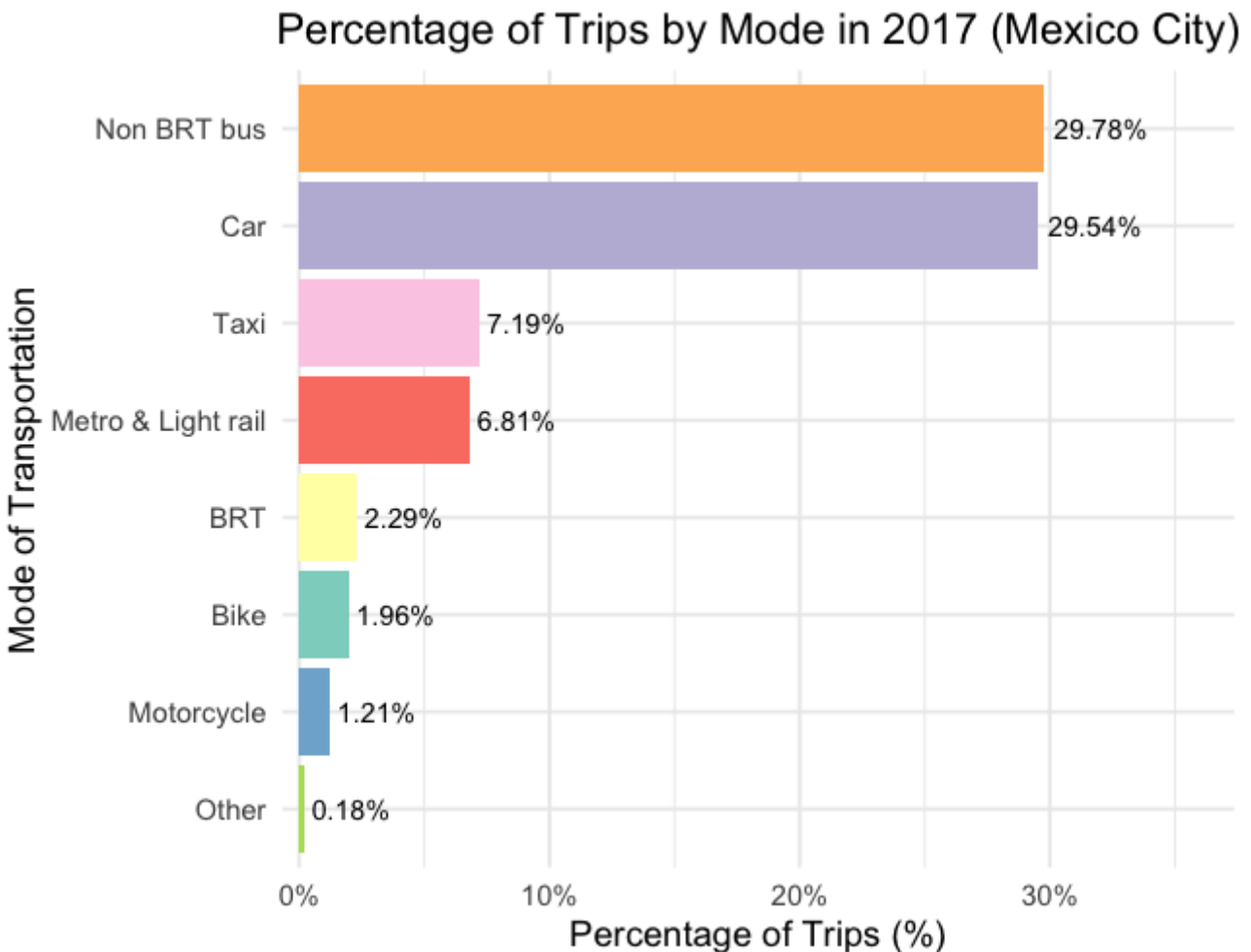
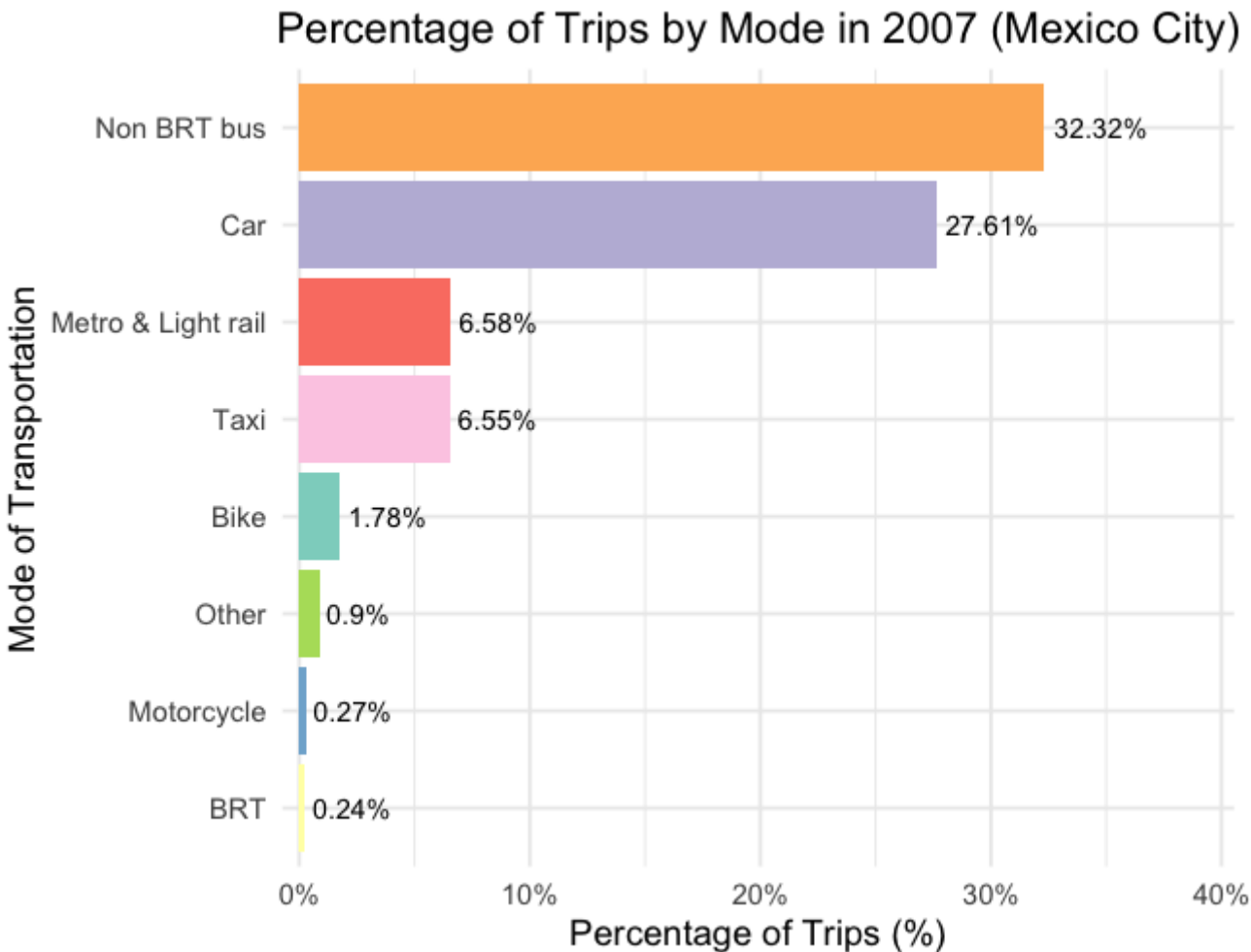
Together, these results demonstrate that while Mexico City's mobility remains bus-dominated with the higher share of multimodal connected trips. These patterns have direct implications for how future investments in infrastructure, integration, and service design can better support sustainable mobility.

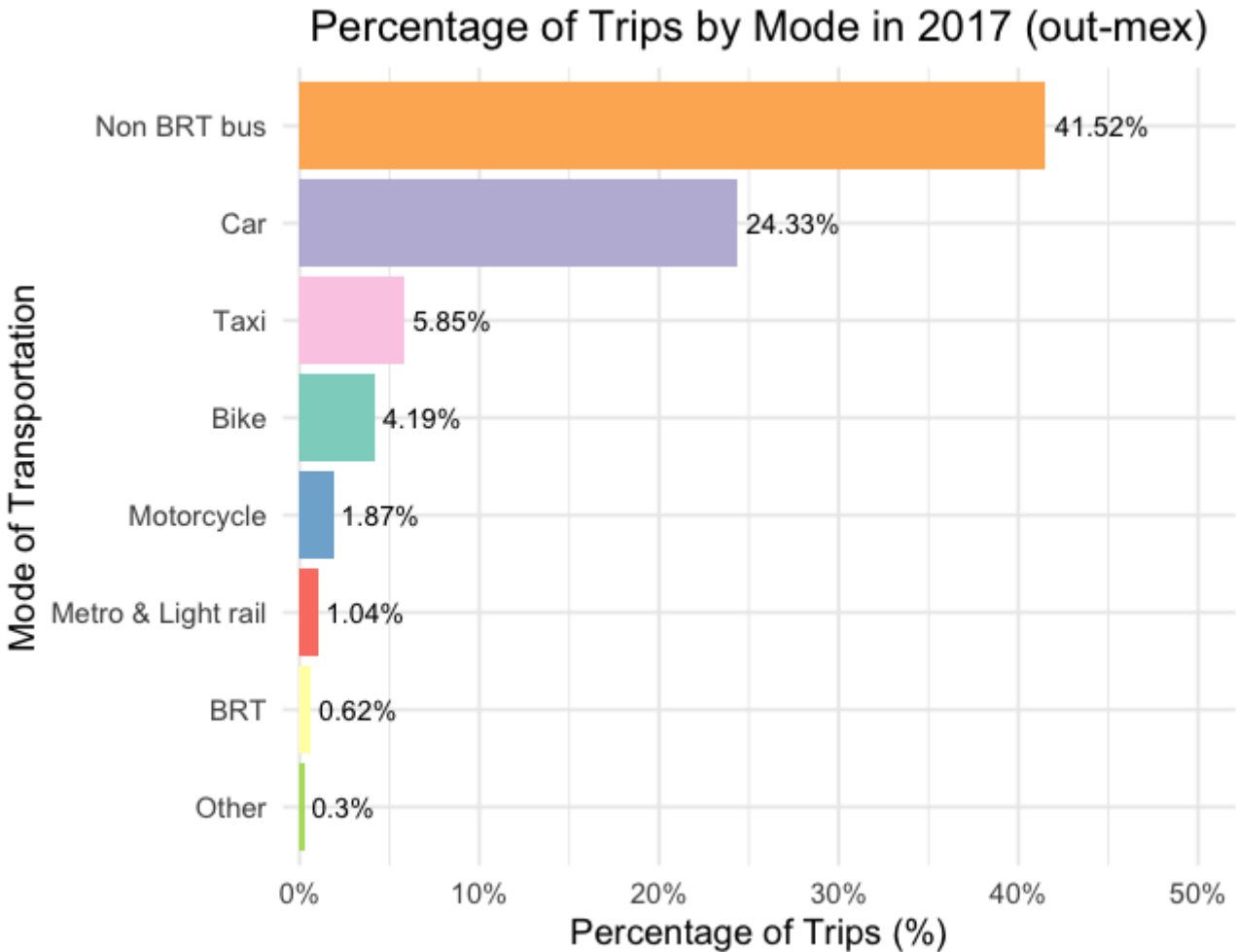
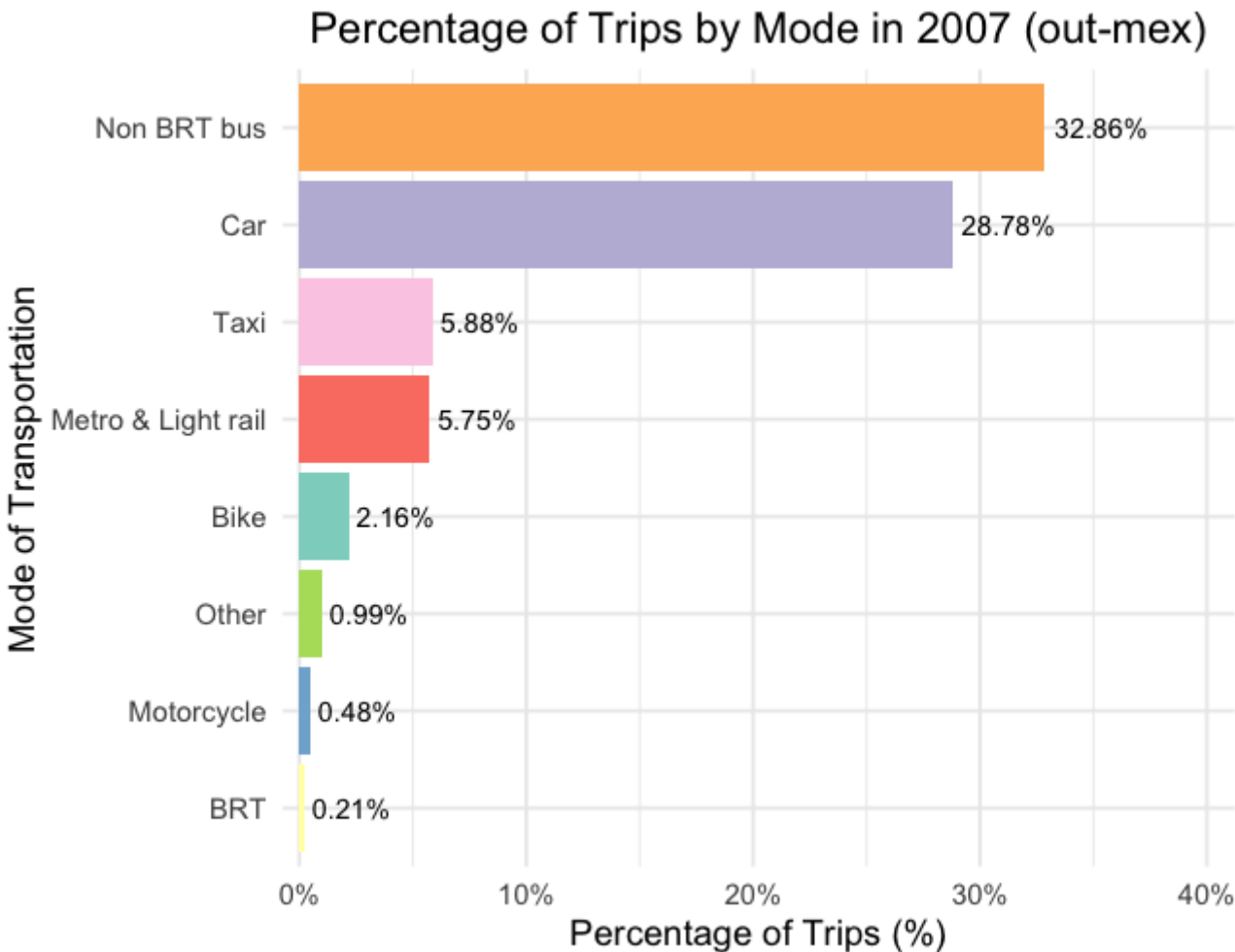
Mexico City Core and Surrounding areas trip Comparison

To understand the differences in travel mode choices between Mexico City's core and its surrounding suburban areas, we divided the dataset into two groups based on their household addresses in 2007 and 2017. The core area includes the **ENT=09**, while the surrounding suburban areas encompass **ENT=15, 13**. In both years, the trips is roughly evenly split between the core and surrounding areas.

Single-mode trips comparison

- **Mexico City core:** From 2007 to 2017, single-mode trips increase from 76.24% to 78.98% (exlucing walking), with non-BRT bus remaining the dominant mode, but decreasing from 32.2% to 29.7%. Automobile usage is the second most common mode, increase from 27.6% to 30%. Single trip metro also increase from 6.5% to 6.8%.
- **Surrounding suburban areas:** Single-mode trips also increased from 77% to around 80% (excluding walking). Non-BRT bus remains the most used mode, increasing from 32.58% to 41.52%. Automobile usage decreased from 28.7% to 24.3%. Metro usage decreased from 5.7% to 1.0%.



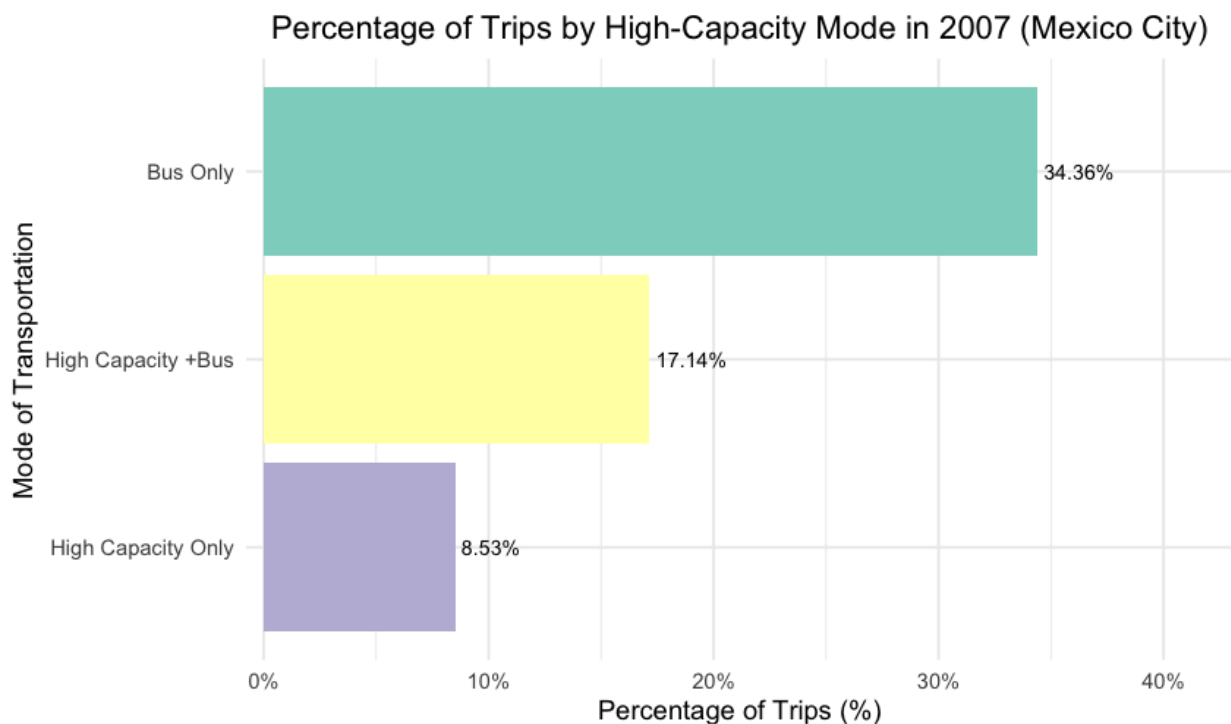


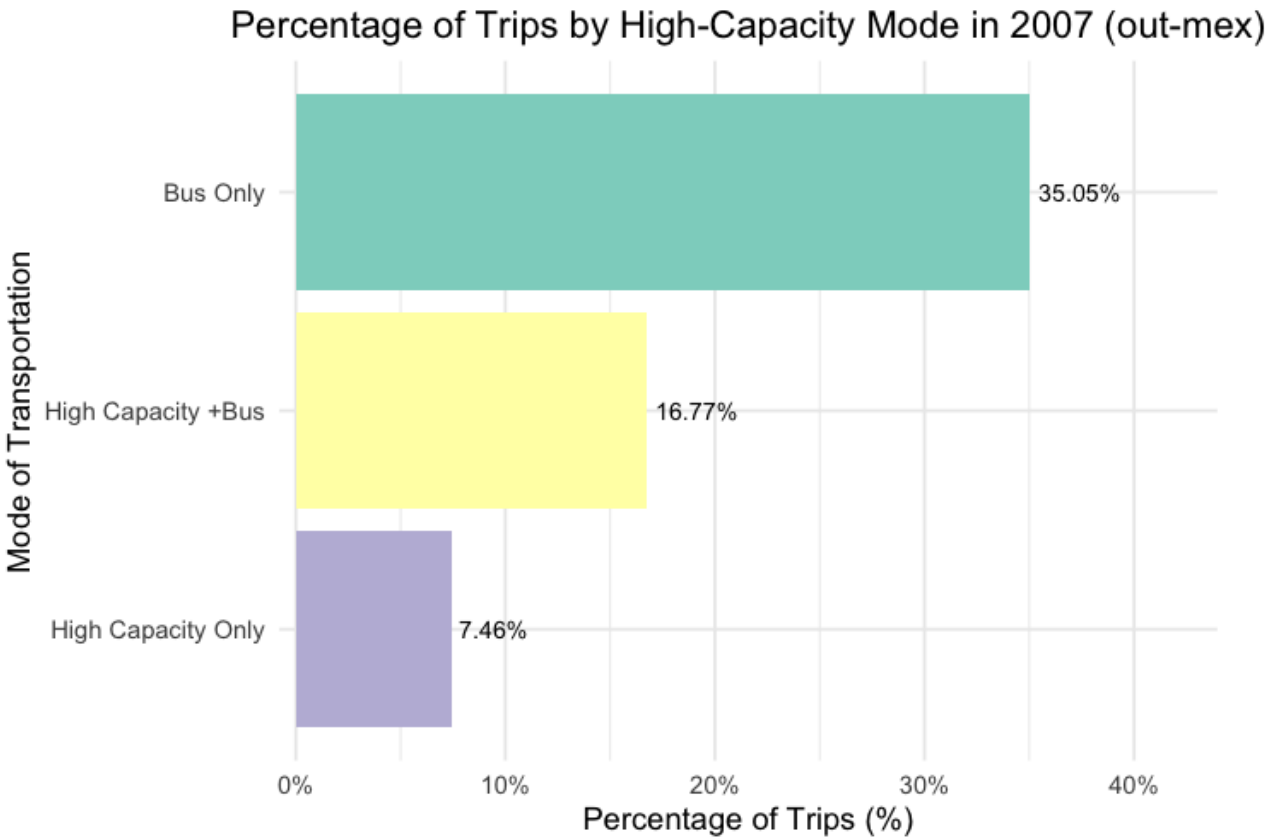
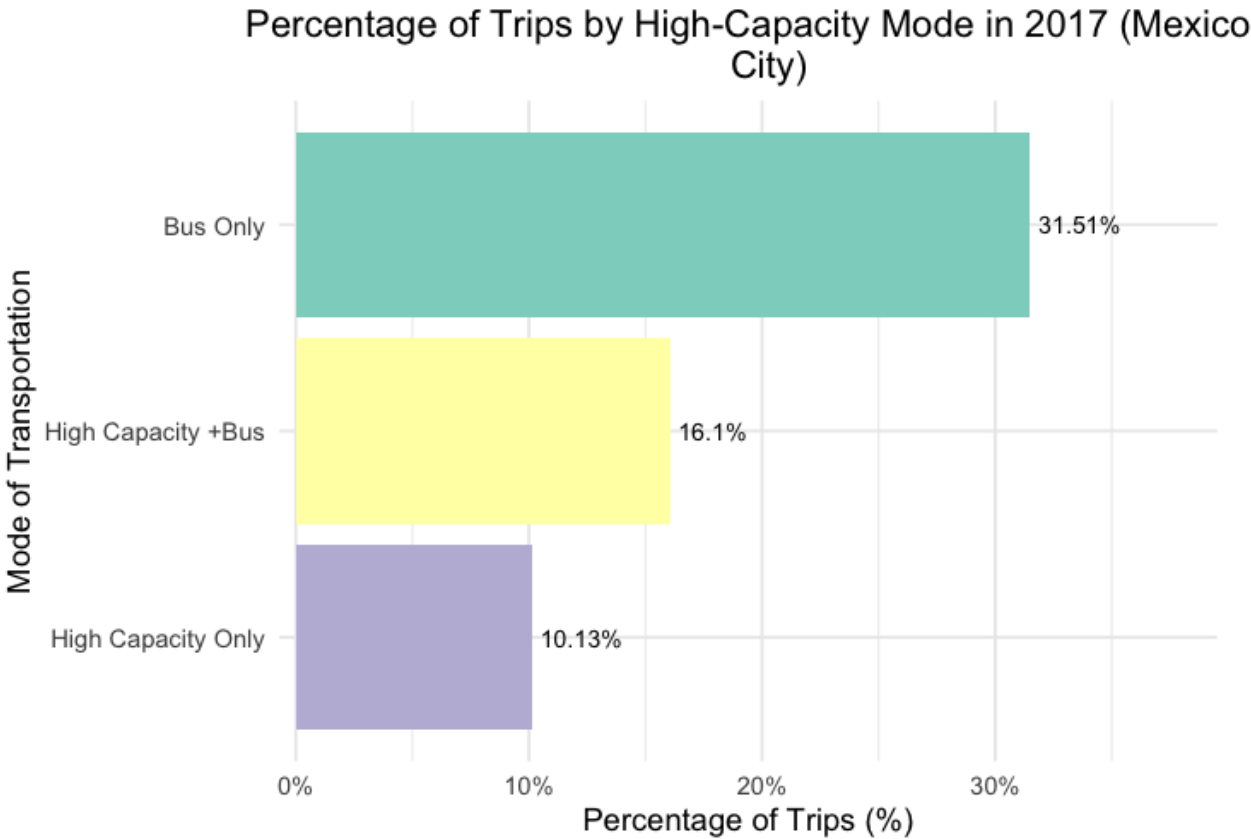
Multimodal trips comparison

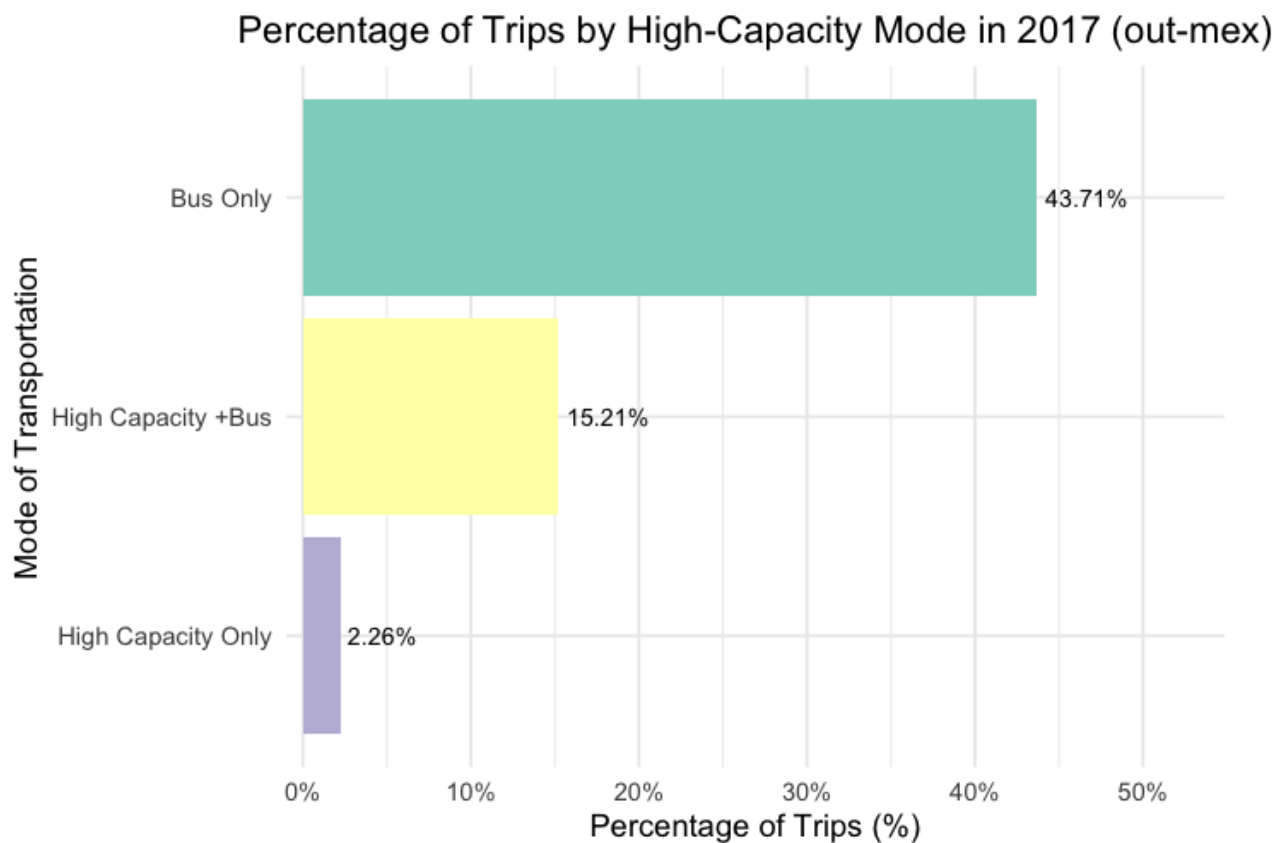
There is no dramatic difference in multimodal trips between Mexico City core and surrounding suburban areas and the overall trends listed in above section. However, the core area has a slightly higher share of multimodal trips involving high-capacity transit (Metro/BRT) compared to the surrounding areas. Multimodal trips share generally decrease over years in both areas.

High Capacity Transit + Bus connectivity comparison

- **Mexico City core:** The share of bus only and bus+high-capacity trips decreased from 2007 to 2017, while high-capacity only trips increased slightly (from 8.53% to 10.13%), which potentially suggests improved accessibility to high-capacity transit in the core area. The share of high-capacity transit + bus trips decreased from 17.14% to 16.1%.
- **Surrounding suburban areas:** The surrounding suburban areas saw a completely different trend. The share of bus only trips increased dramatically from 35% to 43%, while high-capacity only trips decreased from 7.4% to 2.2%. The share of high-capacity transit + bus trips stayed relatively stable, slightly decreasing from 16.77% to 15.21%.







Key takeaways and insights

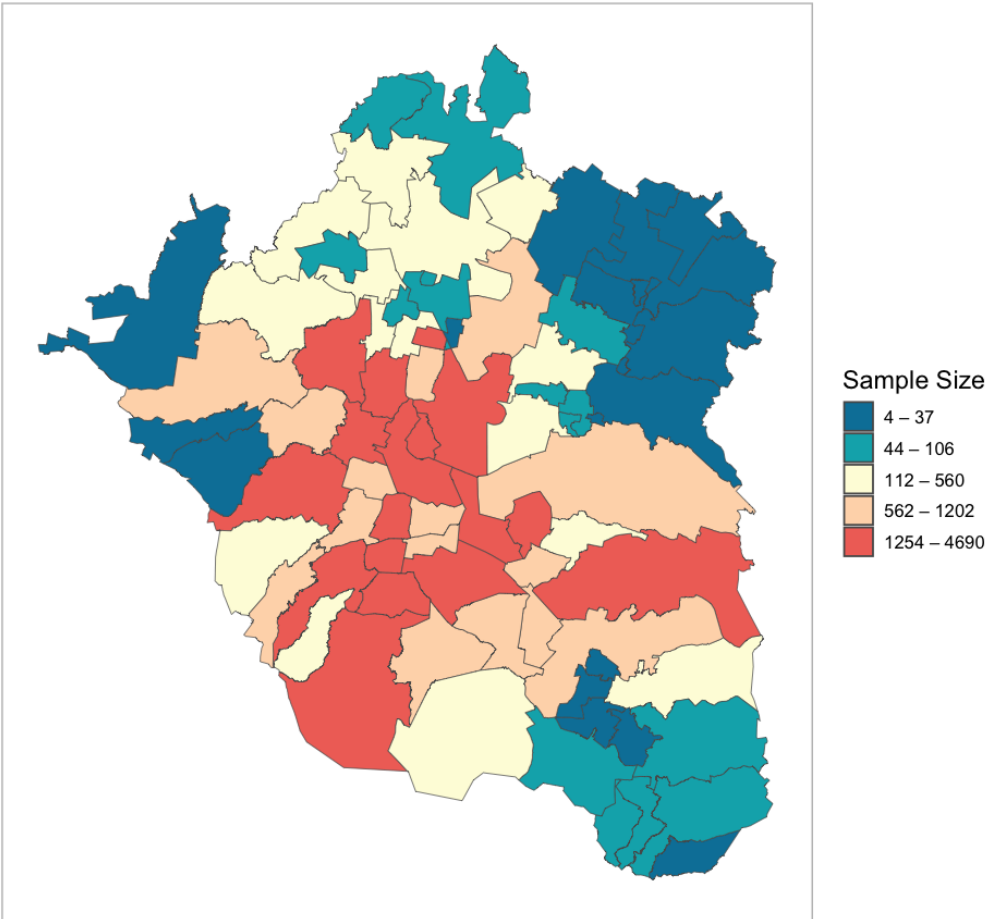
- **Single-mode trips increased** in both Mexico City core and surrounding suburban areas.
- **Non-BRT bus remains the dominant mode** in both areas, with significant growth in suburban areas.
- **High-capacity transit (Metro/BRT) usage decreased in surrounding suburban areas.**

Sampling Examination

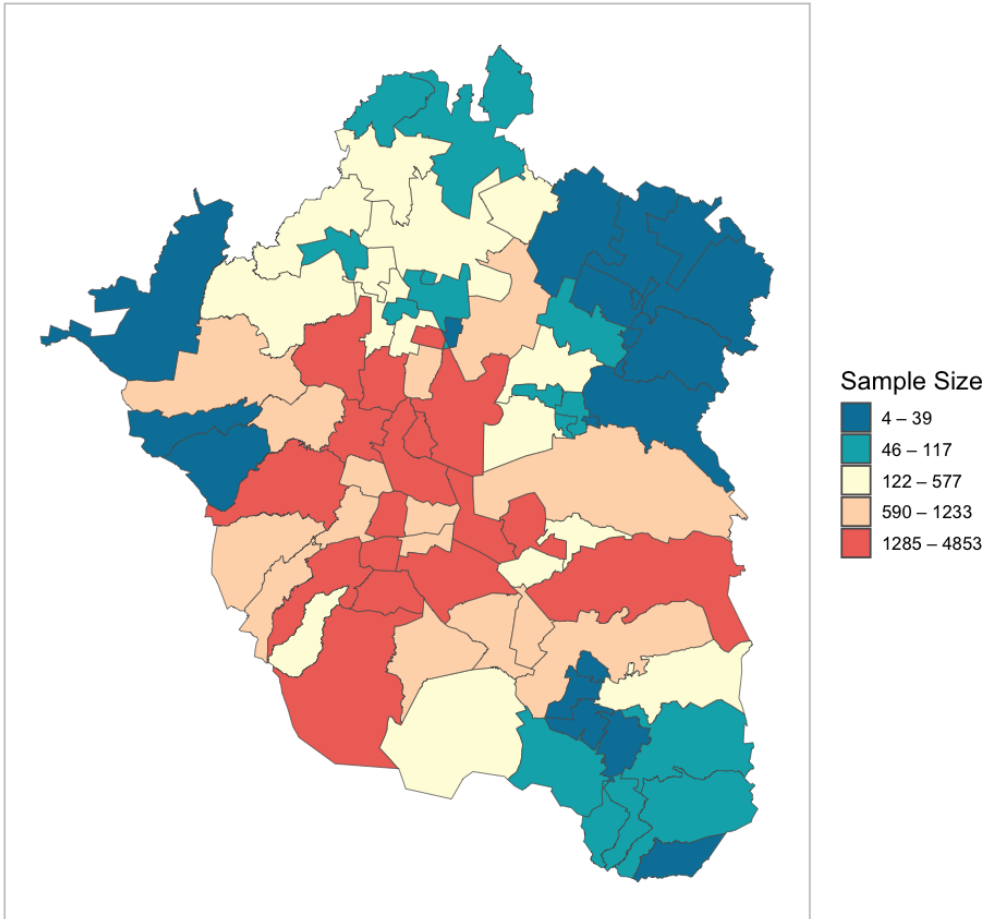
To ensure the reliability of our analysis, we examined the sampling method used in both the 2007 and 2017 Mexico City Household Travel Surveys. We examined the sampling from 2 perspectives: spatial distribution of sampling households at municipality level and trips categories distribution at municipality level.

Spatial distribution of sampling households at municipality level

Sample size of the Survey (2007)



Sample size of the Survey (2017)



Above two maps show the spatial distribution of sampled households in 2007 and 2017 at the municipality level. The color gradient represents the number of sampled households in each municipality, with red colors indicating a higher number of samples, and green represents a smaller number of sample. The break down of color gradient is based on quantile.

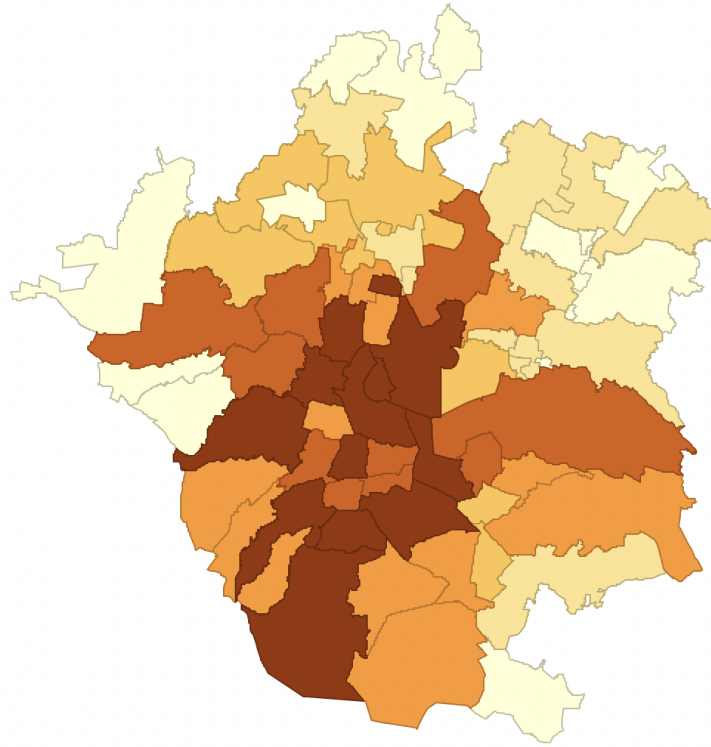
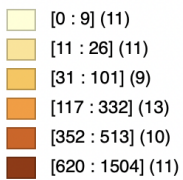
Overall, the spatial distribution of sampled households is quite similar between 2007 and 2017. The core area of Mexico City (central municipalities) has a higher concentration of sampled households in both years, while the surrounding suburban areas have fewer samples. This consistency in spatial distribution suggests that the sampling suggests that the spatial distribution of sample household isn't a contributing factors to the observed changes in travel behavior between 2007 and 2017.

Trips categories distribution at municipality level

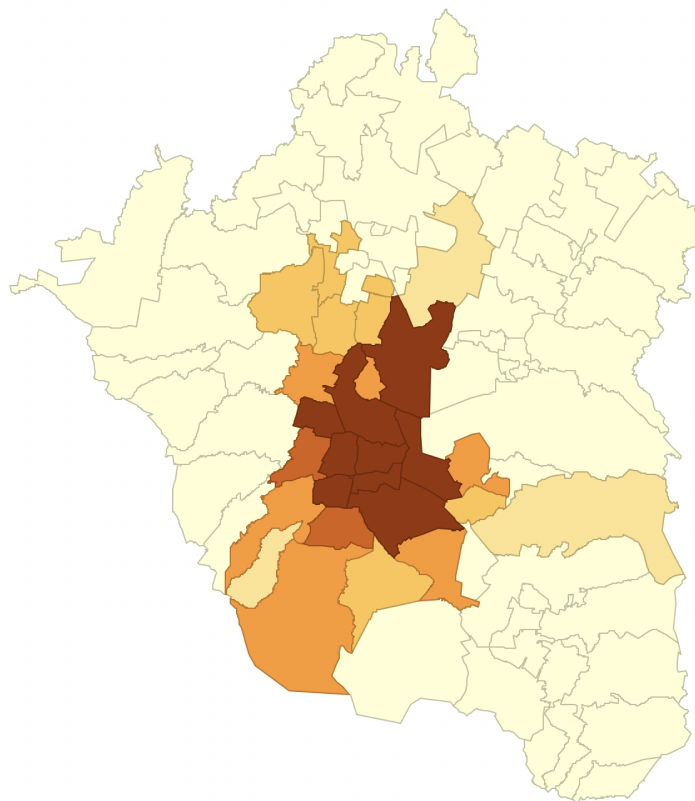
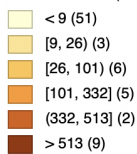
To further validate the sampling method, we examined the distribution of trip categories (high-capacities only, bus only, and bus + high capacities) at the municipality level for both years. The following maps illustrate the trip number in each category across different municipalities, over time.

High-capacities only trips

Quantile: metro_only_col



Custom Breaks (metro_o

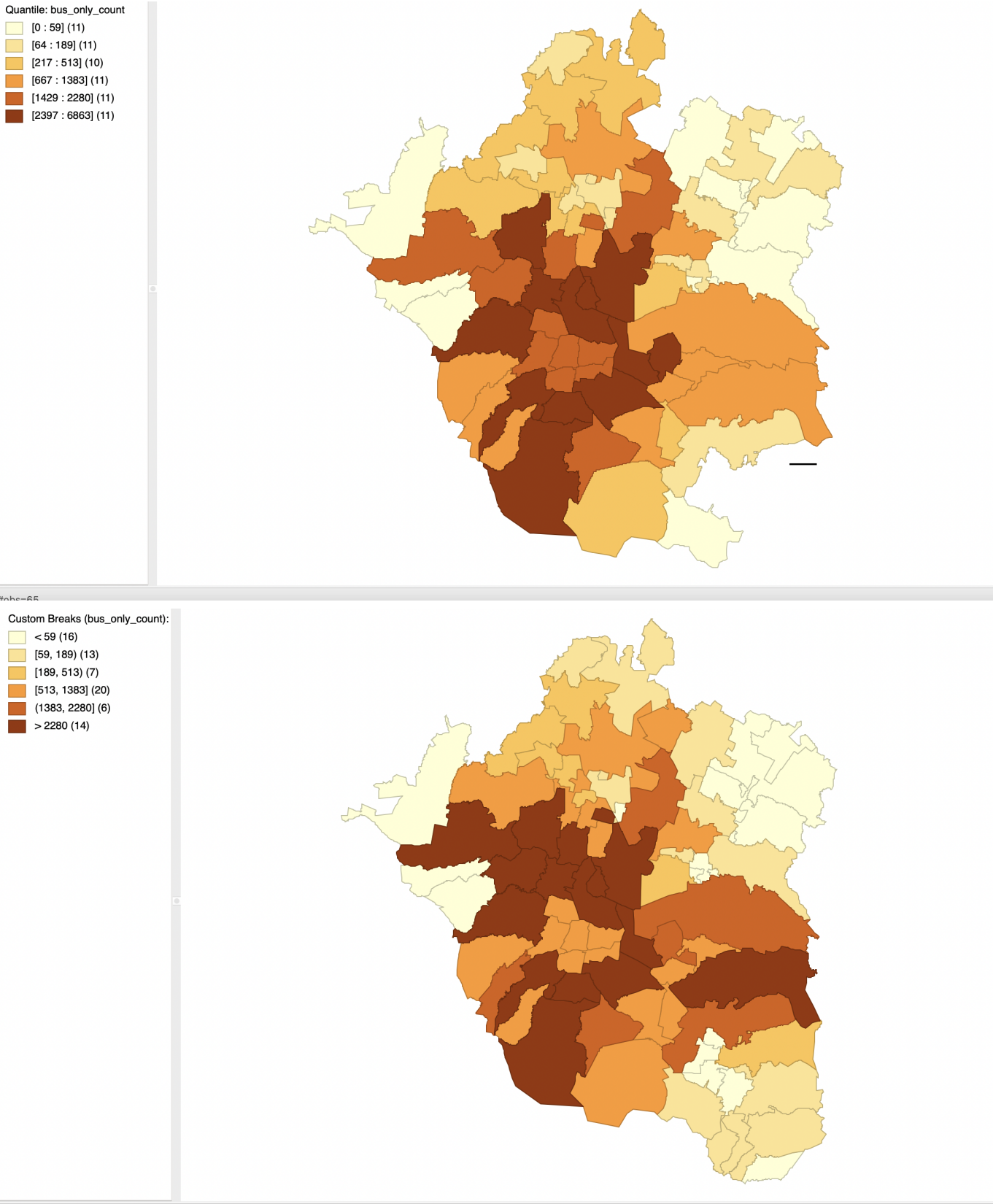


The above maps show the distribution of high-capacities only trips in 2007 and 2017. The color gradient represents the number of high-capacities only trips in each municipality, with higher numbers indicated by dark red colors and lower numbers by light yellow colors. The break down of color gradient is based on quantile (`ntile=6` in 2007, and matching break down in 2017).

Overall, there is a noticeable differences in trip distribution between 2007 and 2017. There is dramatic decrease in high-capacities only trips in most municipalities, especially in the suburban areas. This trends

may indicate a shift in travel behavior, but more likely caused by sampling bias.

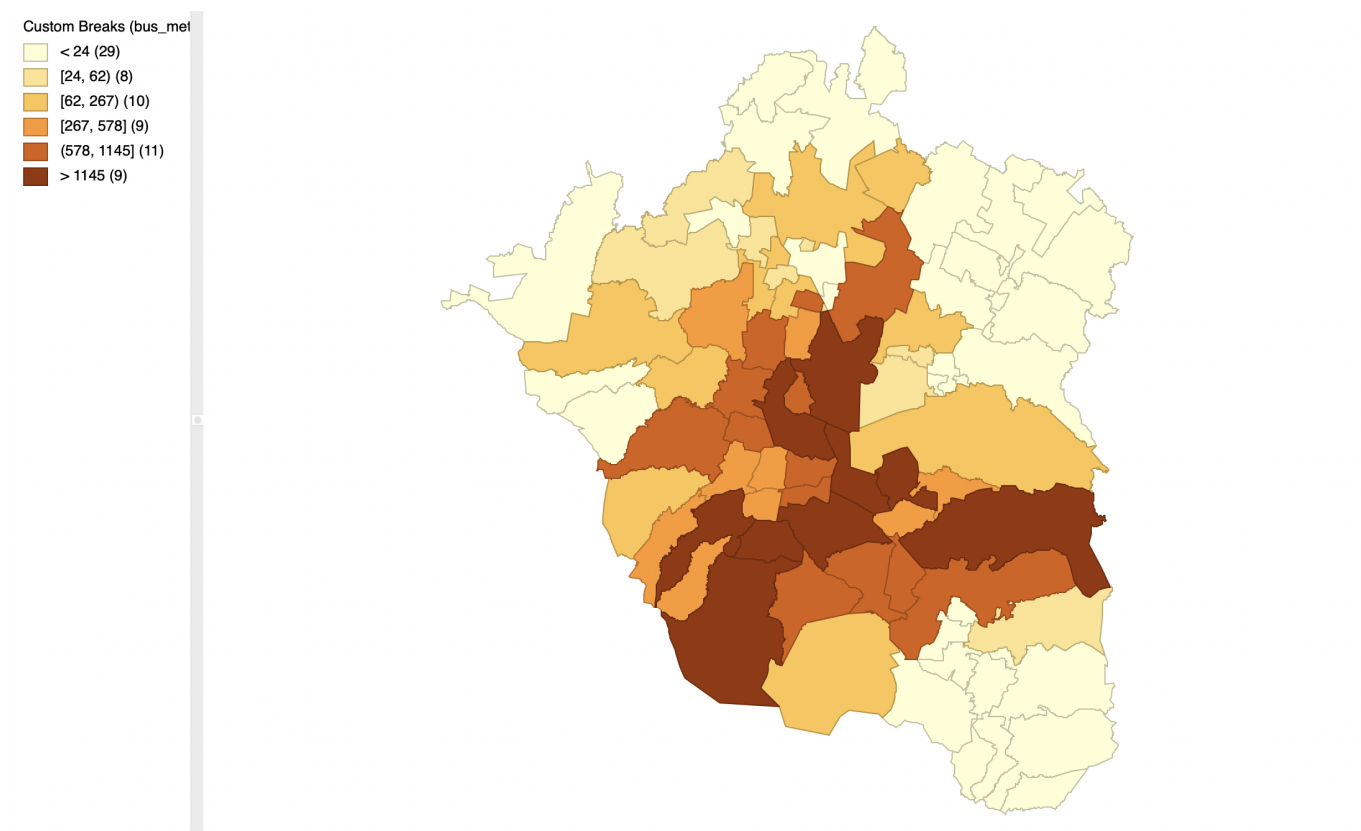
Bus only trips



The above maps show the distribution of bus only trips in 2007 and 2017. The color gradient represents the number of bus only trips in each municipality, with higher numbers indicated by dark red colors and lower numbers by light yellow colors. The break down of color gradient is based on quantile (`ntile=6` in 2007, and matching break down in 2017).

Like the high capacities only trips, there is some differences in bus only trip distribution between 2007 and 2017, but less dramatic. There is a slight increase in bus only trips in most municipalities, especially in the suburban areas on the west side or east side of Mexico City. This trend may indicate a shift in travel behavior, but more likely caused by sampling bias.

Bus + high capacities trips



The above maps show the distribution of bus + high capacities trips in 2007 and 2017. The color gradient represents the number of bus + high capacities trips in each municipality, with higher numbers indicated by dark red colors and lower numbers by light yellow colors. The break down of color gradient is based on quantile (`ntile=6` in 2007, and matching break down in 2017).

The transfer trips shows some shifted between 2007 and 2017, but the changes are not as dramatic as the high capacities only trips. There is a slight increase in bus + high capacities trips in east side of Mexico City, but a slight decrease in north side of Mexico City. This trend may indicate a shift in travel behavior, but more likely caused by sampling bias.